

2025-11-28 - Servidio (15)

Previous lecture: [2025-11-27 - Valentini \(14\)](#)

Next lecture: [2025-12-04 - Servidio \(16\)](#)

Dissipative (incompressible) MHD

$$\begin{cases} \frac{\partial \bar{U}}{\partial t} = -\bar{U} \cdot \nabla \bar{U} - \frac{\nabla P}{\rho} + \bar{J} \times \bar{B} + \nu \nabla^2 \bar{U} \\ \frac{\partial \bar{B}}{\partial t} = \nabla \times (\bar{U} \times \bar{B}) + \eta \nabla^2 \bar{B} \\ \nabla \cdot \bar{U} = \nabla \cdot \bar{B} = 0 \end{cases}$$

The Energy, Cross Helicity and Magnetic Helicity are defined as

$$E = \left\langle \frac{U^2 + B^2}{2} \right\rangle; \quad H_c = \left\langle \frac{\bar{U} \cdot \bar{B}}{2} \right\rangle; \quad H_m = \left\langle \frac{\bar{a} \cdot \bar{B}}{2} \right\rangle$$

we should get as exercise

$$\frac{d}{dt} E = \frac{d}{dt} \left\langle \frac{U^2 + B^2}{2} \right\rangle = -\eta \langle J^2 \rangle - \nu \langle \omega^2 \rangle$$

$$\frac{d}{dt} H_c = \frac{d}{dt} \left\langle \frac{\bar{U} \cdot \bar{B}}{2} \right\rangle = -\frac{\eta + \nu}{2} \langle \bar{J} \cdot \bar{\omega} \rangle$$

$$\frac{d}{dt} H_m = \frac{d}{dt} \left\langle \frac{\bar{a} \cdot \bar{B}}{2} \right\rangle = -\eta \langle \bar{B} \cdot \bar{J} \rangle$$

where

$$\begin{cases} \bar{J} &= \nabla \times \bar{B} \\ \bar{\omega} &= \nabla \times \bar{U} \end{cases}$$

We demonstrate the third equation

$$\frac{d}{dt} \left\langle \frac{\bar{a} \cdot \bar{B}}{2} \right\rangle = \frac{1}{2} \left\langle \frac{\partial \bar{a} \cdot \bar{B}}{\partial t} \right\rangle = \frac{1}{2} \left\langle \bar{B} \cdot \frac{\partial \bar{a}}{\partial t} + \bar{a} \cdot \frac{\partial \bar{B}}{\partial t} \right\rangle$$

and

$$\frac{\partial \bar{B}}{\partial t} = \nabla \times (\bar{U} \times \bar{B}) + \eta \nabla^2 \bar{B} = \nabla \times [\bar{U} \times \bar{B} - \eta \nabla \times \bar{B}]$$

$$\frac{\partial \bar{a}}{\partial t} = \bar{U} \times \bar{B} - \eta \nabla \times \bar{B} - \nabla \chi$$

from the third one (recall $\bar{B} \perp (\bar{U} \times \bar{B})$)

$$\begin{aligned}
\frac{d}{dt} \left\langle \frac{\bar{a} \cdot \bar{B}}{2} \right\rangle &= \frac{1}{2} \left\langle \bar{B} \cdot \left[\cancel{\bar{U} \times \bar{B}} - \eta \nabla \times \bar{B} - \nabla \chi \right] \right\rangle + \frac{1}{2} \left\langle \bar{a} \cdot \left[\nabla \times (\bar{U} \times \bar{B}) + \eta \nabla^2 \bar{B} \right] \right\rangle = \\
&= -\frac{\eta}{2} \left\langle \bar{B} \cdot (\nabla \times \bar{B}) \right\rangle + \frac{1}{2} \left\langle \nabla \cdot (\bar{B} \chi) \right\rangle + \frac{1}{2} \left\langle \bar{a} \cdot [(\bar{B} \cdot \nabla) \bar{U} - (\bar{U} \cdot \nabla) \bar{B}] \right\rangle + \frac{\eta}{2} \left\langle \bar{a} \cdot \nabla^2 \bar{B} \right\rangle = \\
&= -\frac{\eta}{2} \left\langle \bar{B} \cdot (\nabla \times \bar{B}) \right\rangle + \frac{1}{2} \left\langle \nabla \cdot (\bar{B} \chi) \right\rangle + \frac{1}{2} \left\langle \bar{a} \cdot [\nabla \cdot (\bar{B} \bar{U}) - \nabla \cdot (\bar{U} \bar{B})] \right\rangle + \frac{\eta}{2} \left\langle \bar{a} \cdot \nabla^2 \bar{B} \right\rangle
\end{aligned}$$

where we used the following vector identities

$$\begin{aligned}
\nabla \times (\bar{A} \times \bar{B}) &= \bar{A}(\nabla \cdot \bar{B}) - \bar{B}(\nabla \cdot \bar{A}) + (\bar{B} \cdot \nabla) \bar{A} - (\bar{A} \cdot \nabla) \bar{B} \\
\nabla \cdot (\bar{A} \bar{B}) &= (\nabla \cdot \bar{A}) \bar{B} + (\bar{A} \cdot \nabla) \bar{B}
\end{aligned}$$

from boundary condition we reduce it to

$$\begin{aligned}
\frac{d}{dt} \left\langle \frac{\bar{a} \cdot \bar{B}}{2} \right\rangle &= -\frac{\eta}{2} \left\langle \bar{B} \cdot (\nabla \times \bar{B}) \right\rangle + \frac{1}{2} \left\langle \nabla \cdot (\bar{B} \chi) \right\rangle + \frac{1}{2} \left\langle \nabla \cdot [(\bar{a} \cdot \bar{B}) \bar{U} - (\bar{a} \cdot \bar{U}) \bar{B}] \right\rangle + \frac{\eta}{2} \left\langle \bar{a} \cdot \nabla^2 \bar{B} \right\rangle \\
\frac{d}{dt} \left\langle \frac{\bar{a} \cdot \bar{B}}{2} \right\rangle &= -\frac{\eta}{2} \left\langle \bar{B} \cdot (\nabla \times \bar{B}) \right\rangle + \frac{\eta}{2} \left\langle \bar{a} \cdot \nabla^2 \bar{B} \right\rangle \\
\bar{a} \cdot \nabla^2 \bar{B} &= -\nabla \cdot [(\nabla \times \bar{B}) \times \bar{a}] - (\nabla \times \bar{B}) \cdot (\nabla \times \bar{a}) \\
\frac{d}{dt} \left\langle \frac{\bar{a} \cdot \bar{B}}{2} \right\rangle &= -\frac{\eta}{2} \left\langle \bar{J} \cdot \bar{B} \right\rangle + \frac{\eta}{2} \left\langle \nabla \cdot [(\nabla \times \bar{B}) \times \bar{a}] \right\rangle - \frac{\eta}{2} \left\langle \underbrace{(\nabla \times \bar{B})}_{\bar{J}} \cdot \underbrace{(\nabla \times \bar{a})}_{\bar{B}} \right\rangle
\end{aligned}$$

the second term goes away due to boundary conditions.

Finally

$$\frac{d}{dt} H_m = \frac{d}{dt} \left\langle \frac{\bar{a} \cdot \bar{B}}{2} \right\rangle = -\eta \langle \bar{J} \cdot \bar{B} \rangle$$

DONE

So finally

$$\begin{aligned}
\frac{d}{dt} \left\langle \frac{U^2 + B^2}{2} \right\rangle &= -\nu \langle \omega^2 \rangle - \eta \langle J^2 \rangle \\
\frac{d}{dt} \left\langle \frac{\bar{U} \cdot \bar{B}}{2} \right\rangle &= -\frac{\eta + \nu}{2} \langle \bar{\omega} \cdot \bar{J} \rangle \\
\frac{d}{dt} \left\langle \frac{\bar{a} \cdot \bar{B}}{2} \right\rangle &= -\eta \langle \bar{B} \cdot \bar{J} \rangle
\end{aligned}$$

Nuclear reactor or star

Equilibria, everything very slow, energy in mag field
(also in the sun)

Magnetostatic equilibria

We consider the energy with $\bar{U} = 0$

$$\frac{d}{dt} \left\langle \frac{B^2}{2} \right\rangle = -\eta \langle J^2 \rangle$$

we use Fourier transform (we leave behind the norm constant)

$$\bar{B}(\bar{x}, t) = \sum_{\bar{k}} \tilde{B}(\bar{k}, t) e^{i\bar{k} \cdot \bar{x}}, \quad \tilde{B}_j(\bar{k}, t) \in \mathbb{C}$$

eg.

$$B_x(x, y, z, t) = \sum_{k_x=-\infty}^{\infty} \sum_{k_y=-\infty}^{\infty} \sum_{k_z=-\infty}^{\infty} \tilde{B}_x(k_x, k_y, k_z, t) e^{i(k_x x + k_y y + k_z z)}$$

and

$$\tilde{B}(\bar{k}, t) = \frac{1}{V} \int \bar{B}(\bar{x}, t) e^{-i\bar{k} \cdot \bar{x}} d^3x = \langle \bar{B}(\bar{x}, t) e^{-i\bar{k} \cdot \bar{x}} \rangle$$

We need to transform B^2

$$B^2 = \bar{B} \cdot \bar{B} = \left(\sum_{\bar{k}} \tilde{B}(\bar{k}, t) e^{i\bar{k} \cdot \bar{x}} \right) \cdot \left(\sum_{\bar{l}} \tilde{B}(\bar{l}, t) e^{i\bar{l} \cdot \bar{x}} \right)$$

so

$$\frac{1}{2} \frac{\partial B^2}{\partial t} = \frac{\partial}{\partial t} \sum_{\bar{k}} \sum_{\bar{l}} \frac{\tilde{B}(\bar{k}, t) \tilde{B}(\bar{l}, t)}{2} e^{i(\bar{k}+\bar{l}) \cdot \bar{x}}$$

and the energy can be written as

$$\begin{aligned} \frac{d}{dt} \left\langle \frac{B^2}{2} \right\rangle &= \frac{d}{dt} \frac{1}{V} \sum_{\bar{k}} \sum_{\bar{l}} \int \frac{\tilde{B}(\bar{k}, t) \tilde{B}(\bar{l}, t)}{2} e^{i(\bar{k}+\bar{l}) \cdot \bar{x}} d^3x = \\ &= \frac{d}{dt} \sum_{\bar{k}} \sum_{\bar{l}} \frac{\tilde{B}(\bar{k}, t) \tilde{B}(\bar{l}, t)}{2} \frac{1}{V} \int e^{i(\bar{k}+\bar{l}) \cdot \bar{x}} d^3x \end{aligned}$$

We use the orthonormality condition in the integral

$$\frac{1}{V} \int e^{i(\bar{k}+\bar{l}) \cdot \bar{x}} d^3x = \delta_{\bar{k}, -\bar{l}}$$

so

$$\begin{aligned} \frac{d}{dt} \frac{\langle B^2 \rangle}{2} &= \frac{d}{dt} \sum_{\bar{k}} \sum_{\bar{l}} \frac{\tilde{B}(\bar{k}, t) \tilde{B}(\bar{l}, t)}{2} \delta_{\bar{k}, -\bar{l}} = \\ &= \frac{d}{dt} \sum_{\bar{k}} \frac{\tilde{B}(\bar{k}, t) \tilde{B}(-\bar{k}, t)}{2} = \\ &= \frac{d}{dt} \sum_{\bar{k}} \frac{|\tilde{B}(\bar{k}, t)|^2}{2} \end{aligned}$$

where we used the reality condition ($\bar{B}$ real, Fourier transformation of $\bar{B}$ is symmetric)

$$f \in \mathbb{R} \quad \Longrightarrow \quad f(k) = f(-k)$$

We can do the same reasoning also for $\bar{J}$ in order to find J^2 .

$$\bar{J} = \nabla \times \bar{B} = \nabla \times \left[\sum_{\bar{k}} \tilde{\bar{B}}(\bar{k}, t) e^{i\bar{k} \cdot \bar{x}} \right] = \sum_{\bar{k}} i[\bar{k} \times \tilde{\bar{B}}(\bar{k}, t)] e^{i\bar{k} \cdot \bar{x}}$$

so

$$J^2 = \sum_{\bar{k}} \sum_{\bar{l}} i^2 [\bar{k} \times \tilde{\bar{B}}(\bar{k}, t)] \cdot [\bar{l} \times \tilde{\bar{B}}(\bar{l}, t)] e^{i(\bar{k} + \bar{l}) \cdot \bar{x}}$$

and

$$\langle J^2 \rangle = -\frac{1}{V} \int \sum_{\bar{k}} \sum_{\bar{l}} [\bar{k} \times \tilde{\bar{B}}(\bar{k}, t)] \cdot [\bar{l} \times \tilde{\bar{B}}(\bar{l}, t)] e^{i(\bar{k} + \bar{l}) \cdot \bar{x}} d^3x = - \sum_{\bar{k}} |\bar{k} \times \tilde{\bar{B}}(\bar{k}, t)|^2$$

We notice that

$$\nabla \cdot \bar{B} = 0 \quad \Longrightarrow \quad \bar{k} \cdot \tilde{\bar{B}}(\bar{k}, t) = 0 \quad \Longrightarrow \quad \bar{k} \perp \tilde{\bar{B}}(\bar{k}, t)$$

so

$$\langle J^2 \rangle = - \sum_{\bar{k}} k^2 |\tilde{\bar{B}}(\bar{k}, t)|^2 = -2 \sum_{\bar{k}} k^2 E_b(\bar{k}, t)$$

where we used

$$E_b(\bar{k}, t) \equiv \frac{|\tilde{\bar{B}}(\bar{k}, t)|^2}{2}$$

so our original equation can be written as

$$\frac{d}{dt} \left\langle \frac{B^2}{2} \right\rangle = -\eta \langle J^2 \rangle \quad \Longrightarrow \quad \frac{d}{dt} \left[\sum_{\bar{k}} E_b(\bar{k}, t) \right] = -2\eta \sum_{\bar{k}} k^2 E_b(\bar{k}, t)$$

Suppose that energy is stored in few (large) modes

$$\frac{d}{dt} E_b(\bar{k}, t) \simeq -\eta \cdot k^2 E_b(\bar{k}, t)$$

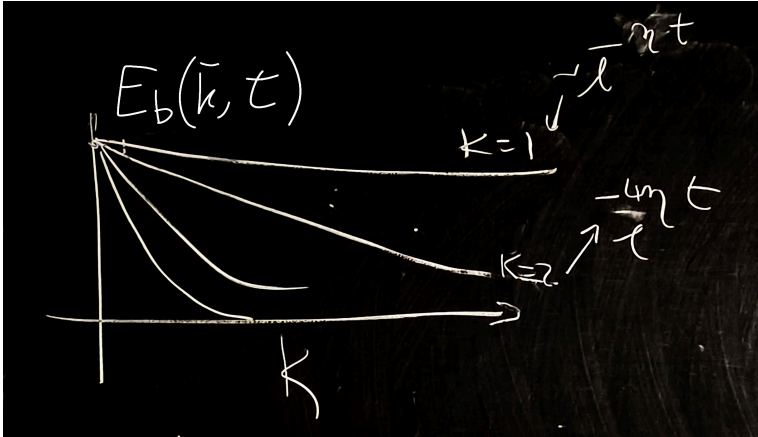
which is an exponential of E_b in time which can be solved by separation of variables

$$\frac{dE_b(\bar{k}, t)}{E_b(\bar{k}, t)} = -\eta \cdot k^2 dt \quad \Longrightarrow \quad E_b(\bar{k}, t) = E_b^0(\bar{k}) e^{-\eta k^2 t}$$

with the **resistive time** defined as

$$\tau_d(k) \equiv \frac{1}{\eta k^2}$$

PIC decaying for rk vectors



Also for fluids we have

$$\frac{d}{dt} \left\langle \frac{U^2}{2} \right\rangle \sim -\nu \omega^2 \quad \Rightarrow \quad |U_{\vec{k}}(\vec{k}, t)|^2 \sim e^{-\nu k^2 t}$$

small vortices die much faster than large vortex.

For the viscosity, the bigger it is, the smaller the dissipation time.

"we cannot create small vortices in jelly "

With $\vec{U} = 0$, $\frac{d}{dt} \left\langle \frac{1}{2} \vec{U} \cdot \vec{B} \right\rangle$ (**Cross Helicity**) does not give us much information.

We have a look at the **Magnetic Helicity** (recall $\nabla \times \vec{a} = \vec{B}$)

$$\frac{d}{dt} H_m = \frac{d}{dt} \left\langle \frac{\vec{a} \cdot \vec{B}}{2} \right\rangle = -\eta \langle \vec{J} \cdot \vec{B} \rangle$$

We notice that

$$\nabla \times \vec{B} = \nabla \times (\nabla \times \vec{a}) = \cancel{\nabla(\nabla \cdot \vec{a})} - \nabla^2 \vec{a}$$

so

$$k^2 \vec{a}(\vec{k}, t) = i \vec{k} \times \vec{\tilde{B}}(\vec{k}, t) \quad \Rightarrow \quad \vec{\tilde{a}}(\vec{k}, t) = i \frac{\vec{k} \times \vec{\tilde{B}}(\vec{k}, t)}{k^2}$$

So the Magnetic Helicity in Fourier space is

$$\begin{aligned} \frac{d}{dt} \sum_{\vec{k}} \frac{\vec{\tilde{a}}(\vec{k}, t) \cdot \vec{\tilde{B}}(\vec{k}, t)}{2} &= -\eta \sum_{\vec{k}} \sum_{\vec{l}} i(\vec{k} \times \vec{\tilde{B}}(\vec{k}, t)) \cdot \vec{\tilde{B}}(\vec{l}, t) e^{i(\vec{k}+\vec{l}) \cdot \vec{x}} \\ \frac{d}{dt} H_m &\sim i\eta (\vec{k} \times \vec{B}(\vec{k}, t)) \cdot \vec{\tilde{B}}(\vec{k}, t) \sim i\eta k (\vec{\tilde{B}}_{\vec{k}}^2) \sim i\eta k E_b(\vec{k}, t) \end{aligned}$$

which is complex and can "bounce", it has a phase, it has a weak dependence on $\vec{k}$.

Magnetic twist will survive for a long time.

Taylor Force Free States

J.B. Taylor, PRL 33, 1139 (1974)

In magnetostatic equilibria ($\vec{U} = 0, \nu = 0, \eta \neq 0$) we have 2 invariants

$$E = \left\langle \frac{B^2}{2} \right\rangle; \quad H_m = \left\langle \frac{\bar{a} \cdot \bar{B}}{2} \right\rangle$$

We will solve a variation minimization problem

$$\delta(E - \mu H_m) = 0 \quad \implies \quad \delta \int \left[\frac{B^2}{2} - \frac{\mu(\bar{a} \cdot \bar{B})}{2} \right] d^3x = 0$$

Using the Euler-Lagrange equations

$$\delta \int_{-\infty}^{\infty} \mathcal{L}(t, x(t), \dot{x}(t)) dt = 0 \quad \implies \quad \frac{\partial \mathcal{L}}{\partial q} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} = 0$$

We can extend it to multidimensional lagrangian

$$\int_{\Omega} \mathcal{L}(x_1, x_2, \dots, x_{n-1}, x_n, f_1, f_2, \dots, f_m, f_{i,j}, \dots) d^d x = 0$$

with

$$f_{i,j} \equiv \frac{\partial f_i}{\partial x_j}$$

Generalized Euler-Lagrange equations

$$\frac{\partial \mathcal{L}}{\partial f_i} - \sum_{j=1}^n \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial f_{i,j}} \right) = 0 \quad \implies \quad \begin{cases} \frac{\partial \mathcal{L}}{\partial f_1} - \sum_{j=1}^n \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial f_{1,j}} \right) = 0 \\ \frac{\partial \mathcal{L}}{\partial f_2} - \sum_{j=1}^n \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial f_{2,j}} \right) = 0 \\ \dots\dots\dots \\ \frac{\partial \mathcal{L}}{\partial f_m} - \sum_{j=1}^n \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial f_{m,j}} \right) = 0 \end{cases}$$

We take f_1 as $\bar{a}$, the lagrangian is

$$\begin{aligned} \mathcal{L} &= \frac{B^2}{2} - \mu \frac{\bar{a} \cdot \bar{B}}{2} = \frac{(\nabla \times \bar{a})^2}{2} - \mu \frac{\bar{a} \cdot (\nabla \times \bar{a})}{2} = \\ &= \frac{(\partial_y a_z - \partial_z a_y)^2}{2} + \frac{(\partial_z a_x - \partial_x a_z)^2}{2} + \frac{(\partial_x a_y - \partial_y a_x)^2}{2} + \\ &\quad - \frac{\mu}{2} a_x (\partial_y a_z - \partial_z a_y) - \frac{\mu}{2} a_y (\partial_z a_x - \partial_x a_z) - \frac{\mu}{2} a_z (\partial_x a_y - \partial_y a_x) \end{aligned}$$

The Euler-Lagrange equation for $i = 1$ (along $\hat{x}$) is

$$\frac{\partial \mathcal{L}}{\partial a_x} - \partial_x \left[\frac{\partial \mathcal{L}}{\partial (\partial_x a_x)} \right] - \partial_y \left[\frac{\partial \mathcal{L}}{\partial (\partial_y a_x)} \right] - \partial_z \left[\frac{\partial \mathcal{L}}{\partial (\partial_z a_x)} \right] = 0$$

First term (we only differentiate a_x) (using $\nabla \times \bar{a} = \bar{B}$)

$$\frac{\partial \mathcal{L}}{\partial a_x} = -\frac{\mu}{2} (\partial_y a_z - \partial_z a_y) = -\frac{\mu}{2} B_x$$

Second term (we don't have anywhere $\partial_x a_x$)

$$\partial_x \left[\frac{\partial \mathcal{L}}{\partial(\partial_x a_x)} \right] = 0$$

Third term (using $\nabla \times \bar{a} = \bar{B}$)

$$\partial_y \left[\frac{\partial \mathcal{L}}{\partial(\partial_y a_x)} \right] = \partial_y \left[-\frac{2(\partial_x a_y - \partial_y a_x)}{2} + \frac{\mu}{2} a_z \right] = \partial_y \left[-B_z + \frac{\mu}{2} a_z \right]$$

Fourth term

$$\partial_z \left[\frac{\partial \mathcal{L}}{\partial(\partial_z a_x)} \right] = \partial_z \left[\frac{2(\partial_z a_x - \partial_x a_z)}{2} - \frac{\mu}{2} a_y \right] = \partial_z \left[B_y - \frac{\mu}{2} a_y \right]$$

We put everything together

$$-\frac{\mu}{2} B_x - 0 - \partial_y \left[-B_z + \frac{\mu}{2} a_z \right] - \partial_z \left[B_y - \frac{\mu}{2} a_y \right] = 0$$

so

$$\begin{aligned} & -\frac{\mu}{2} B_x + \partial_y B_z - \frac{\mu}{2} \partial_y a_z - \partial_z B_y + \frac{\mu}{2} \partial_z a_y = 0 \\ \implies & -\frac{\mu}{2} B_x - \frac{\mu}{2} (\partial_y a_z - \partial_z a_y) + \partial_y B_z - \partial_z B_y = 0 \\ \implies & -\frac{\mu}{2} B_x - \frac{\mu}{2} B_x + \partial_y B_z - \partial_z B_y = 0 \end{aligned}$$

and finally (using $\bar{J} = \nabla \times \bar{B}$)

$$-\mu B_x = -(\partial_y B_z - \partial_z B_y) \equiv J_x$$

from the other equations we get

$$\begin{cases} J_x = \mu B_x \\ J_y = \mu B_y \\ J_z = \mu B_z \end{cases} \implies \bar{J} = \mu \bar{B}$$

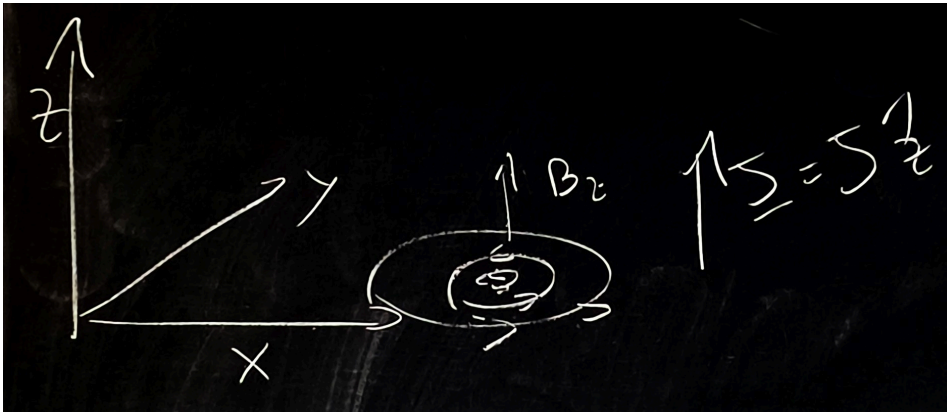
the same reasoning works for $\bar{\omega} = \mu \bar{U}$.

Force-free state, Taylor state

$$\bar{B} \rightarrow \nabla \times \bar{B} = \bar{J} = \mu \bar{B}$$

For example, in tornadoes we have the eye where we find strong upwards winds.

PIC tornado



Generalized Taylor states

$$E = \left\langle \frac{U^2 + B^2}{2} \right\rangle; \quad H_c = \left\langle \frac{\bar{U} \cdot \bar{B}}{2} \right\rangle; \quad H_m = \left\langle \frac{\bar{a} \cdot \bar{B}}{2} \right\rangle$$

with $\nu, \eta \neq 0$

$$\delta \int \left[\left(\frac{U^2 + B^2}{2} \right) - \gamma_1 \frac{\bar{U} \cdot \bar{B}}{2} - \gamma_2 \frac{\bar{a} \cdot \bar{B}}{2} \right] d^3x = 0$$

like Alfvén magnetic field parallel to velocity field.

Solving the minimization problem we get to

$$\begin{cases} \bar{U} = \alpha \bar{B} \\ \nabla \times \bar{B} = \beta \bar{B} \\ \nabla \times \bar{U} = \gamma \bar{U} \end{cases}$$

These equations describe an MHD solution

$$\frac{\partial \bar{U}}{\partial t} = -\bar{U} \cdot \nabla \bar{U} + \bar{J} \times \bar{B} - \nabla P = \cancel{\bar{U} \times \bar{\omega}} + \cancel{\bar{J} \times \bar{B}} - \nabla \left(P + \frac{U^2}{2} \right)$$

where we used

$$\bar{A} \times (\nabla \times \bar{B}) = (\nabla \bar{B}) \cdot \bar{A} - (\bar{A} \cdot \nabla) \bar{B}; \quad \bar{\omega} = \nabla \times \bar{U}$$

Also, using $\bar{U} = \alpha \bar{B}$, we get

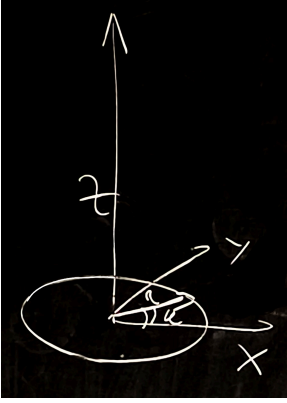
$$\frac{\partial \bar{B}}{\partial t} = \nabla \times (\cancel{\bar{U} \times \bar{B}}) = 0$$

Taylor Force free equilibria in cylindrical coordinates

Taylor states in cylindrical coordinates

$$\bar{B} = (B_r, B_\theta, B_z); \quad B_j = f(r, \theta, z); \quad \nabla \cdot \bar{B} = 0$$

PIC cyl coords



Assumptions:

$$\bar{U} = 0; \quad \nabla \times \bar{B} = \mu \bar{B}$$

so

$$\begin{cases} \nabla \times (\nabla \times \bar{B}) = \nabla(\nabla \cdot \bar{B}) - \nabla^2 \bar{B} = -\nabla^2 \bar{B} \\ \nabla \times (\nabla \times \bar{B}) = \mu \nabla \times \bar{B} = \mu^2 \bar{B} \end{cases}$$

then

$$\nabla^2 \bar{B} + \mu^2 \bar{B} = 0 \quad \implies \quad (\nabla^2 + \mu^2) \bar{B} = 0$$

We isolate the B_z component and use the cylindrical laplacian definition

$$\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2} + \mu^2 \right) B_z = 0$$

We apply the cosine transform over θ and z

$$B_z(r, \theta, z) = \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \tilde{B}_z(r, \theta, z) \cos(m\theta + kz)$$

So the laplacian can be written as

$$\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \mu^2 - k^2 - \frac{m^2}{r^2} \right) \tilde{B}_z(r, m, k) = 0$$

and taking

$$\alpha \equiv \sqrt{\mu^2 - k^2}$$

we get

$$\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \alpha^2 - \frac{m^2}{r^2} \right) \tilde{B}_z(r, m, k) = 0$$

which is the **Bessel differential equation**, which has the following (unique) solution

$$\tilde{B}_z(r, m, k) = J_m(\alpha r)$$

So we can substitute this into the cosine decomposition for B_z

$$B_z(r, \theta, z) = \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} J_m(\alpha r) \cos(m\theta + kz)$$

The other components (B_r and B_θ) are

(Biskamp book has the calculation, not needed for the exam)

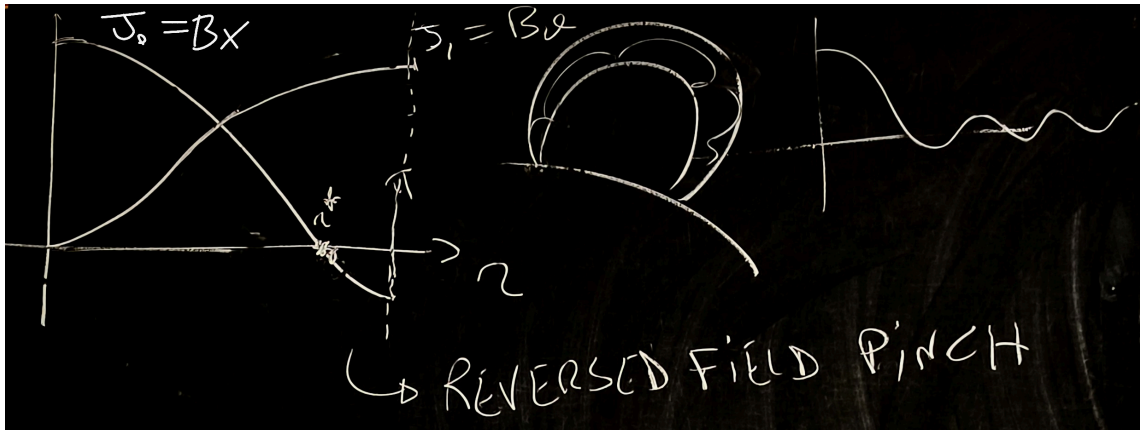
$$B_r(r, \theta, z) = \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \left\{ -\frac{k}{\alpha} J'_m(\alpha r) + \frac{m\mu}{r\alpha^2} J_m(\alpha r) \right\} \sin(m\theta + kz)$$

$$B_\theta(r, \theta, z) = \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \left\{ -\frac{k}{\alpha} J'_m(\alpha r) + \frac{mk}{r\alpha^2} J_m(\alpha r) \right\} \cos(m\theta + kz)$$

We consider only $m = k = 0$ so we have (using $J'_m = J_{m+1}$)

$$\begin{cases} B_z \sim J_0(\mu r) \\ B_r \sim 0 \\ B_\theta = J_1(\mu r) \end{cases}$$

PIC shape of the mag field (1), CME (2), bessel functions (3)



Previous lecture: [2025-11-27 - Valentini \(14\)](#)

Next lecture: [2025-12-04 - Servidio \(16\)](#)